

Branch: Electronics Engineering / Electronics and Communication Engineering /
Electronics and Telecommunication Engineering

Subject Code & Name: BTEXPE702D / BTETPE702D Fiber Optic Communication

Max Marks: 60

Date:03/07/2024

Duration: 3Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/ Marks
CO)

- Q. 1 Solve Any Two of the following.** **12**
- A) Describe the Total Internal Reflection in optical fiber communication. Level 1 6
- B) Describe the phenomenon of refraction. Level 2 6
- C) A light ray travels from air (refractive index $n_1=1$) to water (refractive index $n_2=1.33$). If the angle of incidence is 30 degrees, calculate the angle of refraction. Level 2 6
- Q.2 Solve Any Two of the following.** **12**
- A) What are the main steps involved in the fabrication of optical fibers? Explain any one method of fabrication. Level 1 6
- B) Explain the characteristics and applications of single-mode optical fibers. Level 1 6
- C) Explain how OTDR can detect faults, measure length, and determine attenuation in optical fibers. Level 1 6
- Q. 3 Solve Any Two of the following.** **12**
- A) Describe the basic operating principles of an LED (Light Emitting Diode). How does it generate light? Level 1 6
- B) Describe the operation of an Avalanche Photodiode (APD). Level 2 6
- C) Describe the main components of an optical receiver and their functions. Level 2 6
- Q.4 Solve Any Two of the following.** **12**
- A) Explain the Raman amplification process. Level 1 6
- B) Describe the importance of population inversion and the role of the pump laser in an EDFA. Level 2 6
- C) Describe the architecture of a WDM network. What are the key components and their functions? Level 2 6
- Q. 5 Solve Any Two of the following.** **12**

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| A) Define the different types of nonlinear effects that can occur in fiber optic communication. Explain any one | Level 1 | 6 |
| B) Explain the concept of group velocity dispersion (GVD) and how it affects the propagation of optical pulses in fibers. | Level 1 | 6 |
| C) Define optical solitons and explain how they form in optical fibers. | Level 1 | 6 |

***** End *****